

Study on the Path Planning Based on A* Algorithm for Vascular Intervention Robots

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Abstract - In vascular interventional surgery, accurate path planning is essential to improve surgical success and reduce risk. This paper aimed to verify the effectiveness and practicability of the A* algorithm in the shortest path planning of vascular interventional surgical robot systems. Computed tomography angiography (CTA) vascular images were processed in advance to extract features and calibrate obstacles, simulating the path planning problem in vascular interventional surgery. The A* algorithm was then employed to search for the shortest path from the starting point to the endpoint. Based on the planned path, assistance could be provided to surgeons during operative procedures. The experimental results indicated that the A* algorithm effectively navigated obstacles and identified the shortest path. Moreover, through user-interactive design, this paper offered an intuitive operational experience, allowing users to interactively choose start and end points and display the algorithm's search process and outcomes in real-time. This paper demonstrated the potential application of the A* algorithm within vascular structures and laid the groundwork for the future development of more efficient and intelligent systems for planning and navigating vascular interventional surgeries.

Index Terms - A* algorithm. Path planning. Vascular intervention robots.

I. INTRODUCTION

In recent years, with the rapid advancement of robotics, vascular interventional robots have become a focal point in the field of vascular intervention surgeries. Such robots facilitate stable, precise, and secure surgical operations through meticulous motion control and image guidance [1]. Path planning for these robots stands as a pivotal challenge in achieving autonomous operation and navigation [2]-[3].

Vascular interventional surgery, performed using minimally invasive techniques through catheters, has achieved significant clinical outcomes in improving the treatment efficacy of cardiovascular and cerebrovascular diseases. It effectively treats cardiovascular diseases such as heart diseases and cerebrovascular accidents, reducing patient suffering and recovery time [4]. Traditional surgical procedures have certain limitations, primarily including the following aspects: (1) Manual operation constraints: Traditional vascular interventional surgery typically relies on manual manipulation of catheters and instruments by

physicians, and slight hand tremors or fatigue can affect the precision and efficacy of the surgery [5]; (2) Procedural risks: Since vascular interventional surgery usually involves the manipulation and movement of catheters within blood vessels, there are inherent risks such as vascular damage, thrombosis formation, or infection, which may increase the surgical risks [6]; (3) Anatomical constraints: The anatomical structure of the human vascular system is complex and diverse, with individual variances and variations, which may increase the difficulty and risks of surgery, especially when dealing with complex or anomalous anatomies [7].

Path planning for the vasculature plays a crucial role in medical image processing and computer-assisted surgery [8]. As medical imaging technology rapidly advances and computational algorithms continuously improve, vascular path planning has emerged as a key technology to enhance the navigation, diagnosis, and treatment of vascular surgeries [9]. Automating the determination of optimal pathways within the vascular network, path planning serves as a valuable tool for physicians, enabling more accurate navigation of the vascular system, disease diagnosis, and surgical route planning [10].

However, vascular path planning faces several challenges. Firstly, the intricate topology of the vascular network, which includes branches, bends, and crossings, complicates path planning [11]. Secondly, medical imaging data may be plagued with noise, artifacts, and incompleteness, challenging the accuracy and reliability of path planning [12]. Furthermore, considerations such as surgical risk, time efficiency, and computational complexity must be factored into the clinical application of such planning [13]-[14].

In response to these challenges, numerous vascular path planning algorithms and techniques have emerged recently [15]. Methods based on graph theory, heuristic search, and machine learning have particularly made significant strides in this domain [16]. By modeling the vascular network and optimizing path selection, these methodologies aim to minimize computational complexity while ensuring path connectivity and delivering precise planning outcomes [17].

This paper established an experimental setup based on MATLAB, starting with the preprocessing of vascular images, followed by semantic segmentation to extract and visualize

image features. Subsequently, the start and endpoint on the vasculature were designated with a mouse click. Utilizing the A* algorithm, the paper demonstrated automatic vascular path planning. The processes of path determination, shortest path length retrieval, and the required time for finding the shortest path were all visualized.

The paper was structured as follows: Section I introduced the research background and current technologies, including the progress of vascular interventional robots and path planning algorithms. Section II elucidated the architecture of the vascular interventional surgical robot. Section III detailed the path planning algorithm developed for the vascular interventional robot and its implementation. Section IV presented experimental results and analysis. Lastly, Section V concluded the paper's findings and explored future research directions.

II. VASCULAR INTERVENTION ROBOT

As illustrated in Fig. 1, the vascular interventional robot system under investigation, integrated with an auxiliary imaging system, primarily comprised three main components: a radiation protection operation console, the vascular interventional robot itself, and an auxiliary cart.

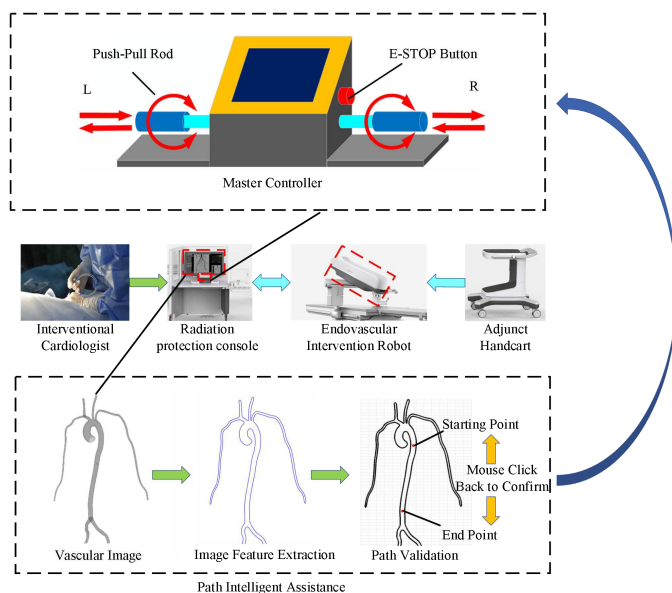


Fig. 1 Image-assisted vascular robot system.

Firstly, the radiation protection console was introduced, which included the main control unit, radiation shielding glass, and a medical information display. The primary function of the medical information display was to provide real-time images from X-rays, CT scans, MRI, etc. Physicians could clearly visualize the blood vessels' structure and the positions of catheters and guide wires. By determining the starting and ending points, a path of the vessels was generated on the display, assisting physicians in manipulating the catheter and guide wire through the main control unit, such as adjusting the catheter's advancement depth or direction. In case of emergencies, an emergency stop button located beside the console was wired in series with the main circuit of the

power supply, enabling direct disconnection of the motor power of the end effector to address sudden events during surgery.

Secondly, the vascular intervention surgical robot was described, whose end-effector arm, guided by remote instructions from the physician, precisely controlled the insertion and movement of the catheter and guide wire. Specific mechanical devices, such as rotation and advancement units, were utilized to adjust the position and depth of the catheter.

Thirdly, the auxiliary trolley was discussed, employed for carrying and transporting various equipment and tools required for surgery, including extra catheters, guide wires, stents, balloon expanders, etc. By organizing and transporting all necessary materials for the operation, the auxiliary trolley reduced the time spent by physicians and operating room staff in fetching materials during surgery, thereby enhancing the overall efficiency of the operating room.

The usage scenario includes: (1) The patient lies on the operating table, covered by imaging equipment such as X-ray machines to ensure real-time image feedback is available. Physicians operate remotely from behind the radiation protection console, manipulating the robotic arm to slowly insert catheters and guide wires into the patient's blood vessels. Real-time digital subtraction angiography (DSA) images are displayed on the monitor at the console, allowing physicians to observe the position and movement of the catheters and guide wires. The basis of the surgery lies in the pre-acquisition of CTA images of the patient's vascular system, which are used to determine the starting and ending points to plan the vascular pathway. This pathway assists physicians in remotely controlling the robot, selecting the correct direction when encountering vascular bifurcations, adjusting the direction of catheters when encountering significant bends in the vessels, or obstacles. (2) Intraoperative monitoring and adjustment stage: If unexpected vascular branches or lesions are encountered, immediate image feedback lets doctors adjust path planning on-the-fly. The imaging system provides essential data for deciding whether to alter intervention strategies or adjust surgical tools. (3) Postoperative evaluation stage: Post-operation, the imaging system checks the stent's position and blood flow to assess the surgery's success. Adjustments can be made based on postoperative imaging, ensuring unimpeded vascular access. Imaging data and robot logs are stored in the patient's electronic health record for future follow-up and management.

III. METHOD

A. Data Set Acquisition and Preprocessing

The data for this paper were obtained from CTA images acquired using computed tomography (CT) technology before vascular interventional surgery. During the CTA scan, patients were typically intravenously injected with contrast agents containing contrast medium. This contrast medium produced a high-density signal in the vascular system, resulting in clear contrast of the blood vessels in the CT images. Subsequently, the CT scanner scanned the patients at different angles using

X-rays, and then the computer combined these scan results into a three-dimensional vascular image, which clearly displayed the structure and blood flow of the vessels. CTA is commonly used to assess vascular diseases such as arterial stenosis, aneurysms, thrombosis, etc., and provided important anatomical information for surgical or interventional treatment.

To extract useful features from the original vascular CTA images, a series of preprocessing steps were employed, as depicted in Fig. 2. The primary objective of the data preprocessing was to enhance the edge information in the images for subsequent feature extraction and analysis. The specific steps included: (1) Grayscale conversion of the images, simplifying the image information to focus more on structure brightness variations. (2) Noise removal and edge detection, involving Gaussian filtering to smooth the grayscale images and remove random noise; calculation of the gradient of the processed images and binarization based on gradient magnitude to highlight the edges in the images. (3) Image thinning and segmentation, which thinned the binarized images and segmented them into independent segments. (4) Extraction and visualization of feature points in the images.

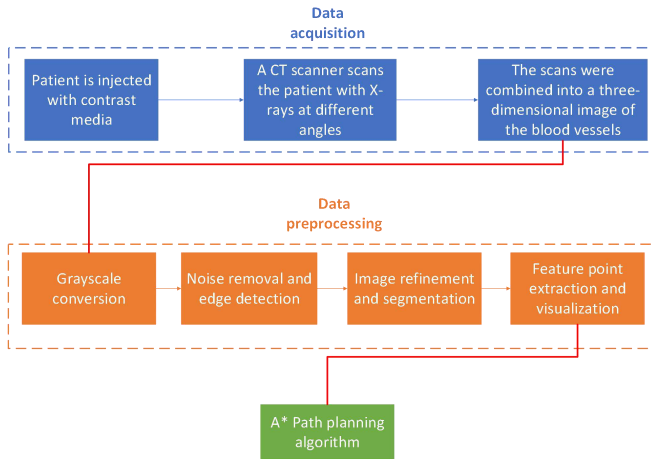


Fig. 2 Steps of data preprocessing.

Through the aforementioned preprocessing steps, key features were successfully extracted from the original vascular images, laying the groundwork for further determination of vascular pathways.

B. Proposed Path Planning A* Algorithm

In the preprocessed images, the A* algorithm was employed for path planning, as shown in Fig. 3. To achieve efficient path planning, this paper utilized the A* algorithm, which is renowned for its effectiveness in path searching and graph traversal problems due to its high efficiency [18]. The A* path planning algorithm finds the shortest path from the initial node to the target node in a graph (e.g., road network, game map). It is considered a heuristic search algorithm because it employs a heuristic function to estimate the image's distance from the current node to the target node [19]. This image's distance is commonly the Euclidean distance from the

current node to the target node or other potential estimation methods [20]-[21].

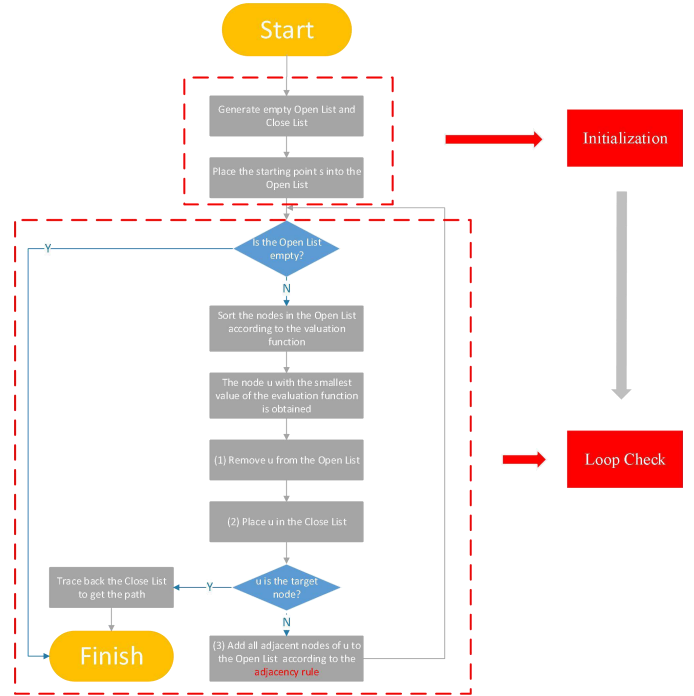


Fig. 3 A* algorithm flow chart.

(1) Valuation function

$$f(x) = g(x) + h(x) \quad (1)$$

Where $f(x)$ is the valuation function in Eq. 1, $g(x)$ represents the actual path cost from the start node to node x ; $h(x)$ denotes the heuristic estimated cost from the current node x to the target node.

(2) The design of $h(x)$

Euclidean Distance: The most common distance measure, defined as the straight-line distance between two points. In two dimensions, the Euclidean distance between two points $P(x_1, y_1)$ and point $Q(x_2, y_2)$, can be calculated using the Pythagorean theorem. It involves considerable computation and is unsuitable for path planning with massive data. the Euclidean Distance d can be expressed as Eq. 2:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (2)$$

Manhattan Distance: Also known as city block distance, it is defined as the sum of the absolute differences of two points' horizontal and vertical coordinates on a standard coordinate system [20]. It simulates the distance of walking from one intersection to another in a city block (excluding diagonal directions). Although this evaluation function does not strictly prioritize direction, it generally ensures that the search direction for the shortest path heads towards the target point. The formula is as follows: For point $P(x_1, y_1)$ and point $Q(x_2, y_2)$, the Manhattan Distance D can be expressed as Eq. 3:

$$D = |x_2 - x_1| + |y_2 - y_1| \quad (3)$$

(3) The flow of the proposed A* algorithm

1. Initialization: Initialize two lists: the Open List, which stores nodes that are accessible but not yet explored, and the Close List, which stores nodes that have been explored (nodes on the optimal path). Add the start node to the Open List.

2. Loop check: As long as the open list is not empty, keep looking. 1) Select the node with the lowest $f(x) = g(x) + h(x)$ value from the open list as the current node. 2) Check whether the current node is the target node. If so, retrace the path and end the search. 3) Remove the current node from the open List and add it to the Close List. 4) For each neighbor of the current node (adjacency rule): If it is already in the closed list, skip it; If it is not in the open list, calculate its $g(x)$, $h(x)$, and $f(x)$, and then add it to the open list; If it is already in the open list, check whether the path to it through the current node is better (i.e., has a lower $g(x)$ value), and if so, update its cost and path.

At the end of the A* algorithm, the lowest cost path from the start node to the destination node is generated. This algorithm is very popular in all kinds of path planning problems, such as game design, robot path planning, and network routing.

C. Implemented of the A* Algorithm

The path planning based on the A* algorithm is shown in Fig. 4. Firstly, the image is preprocessed, and then the path planning based on the A* algorithm is implemented for the processed image, the process is as follows:

(1) Initialization: The starting point is initialized using a heuristic function ($f = g + h$), where g is the cost from the starting point to the current point, and h is the heuristic estimated cost from the current point to the end point (usually the Manhattan distance);

(2) Find neighbors: Find all viable neighbors of the current node. And determine if the candidate is outside the boundary, an obstacle, or has already been explored;

(3) Update and sort: For each neighbor node, calculate its f and g values, add it to the list to be explored according to the f value, and continue to update the list, giving priority to exploring the node with the smallest f value;

(4) Find the shortest path: Repeat the above process until the end point is found, and reconstruct the shortest path from the beginning to the end point by backtracking (using the parent-child relationship of the nodes in the explored list).

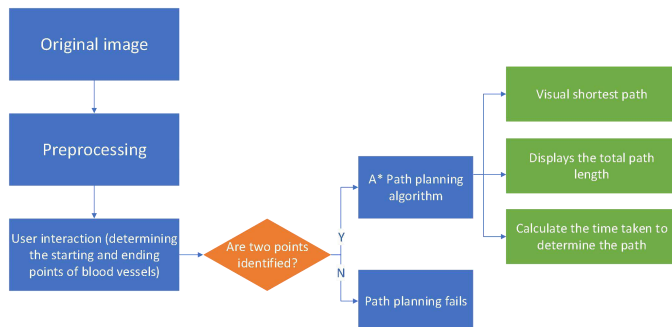


Fig. 4 Path planning based on A* algorithm.

The implementation of the above path planning needs to be accomplished through user interaction (the user sets the start and end point by clicking on the mouse, listens to the click event on the canvas, and records the coordinates of the start and end point, respectively according to the click order). Based on the above results display, after the user sets the starting point and end point, the path search is performed on the image, and the search process is displayed in real-time. Once the shortest path is found, it is plotted on the image, and the total path length and calculation time are displayed.

IV. EXPERIMENTAL RESULTS

A. Experiment Setting

The purpose of this study was to simulate the navigation process of vascular interventional surgery using the A* algorithm for path planning of vascular images. The experimental environment and settings were as follows:

(1) Experimental environment: The environment for this experiment was MATLAB, in which the necessary libraries and toolkits were installed, especially those supporting graph drawing and matrix operations;

(2) Image data: The experiment utilized a blood vessel image obtained through specific imaging technology, rich in detail to represent the vascular structure encountered in actual vascular interventional surgery;

(3) Image preprocessing: The image was first converted into grayscale format and preprocessed through Gaussian filtering and gradient calculation to extract the edge information of blood vessels. On this basis, features of blood vessel images were extracted;

(4) Initialization of Map and Obstacles: The experiment required a predefined grid map encompassing both navigable areas and obstacles, taking the form of a two-dimensional array. Boundaries of the map along with specific areas (features extracted from vascular images) were designated as obstacles. This approach provided flexibility to accommodate various map layouts and obstacle configurations, enabling the identification of the optimal path from the starting point to the destination within complex environments;

(5) User interaction: The experiment provided user interaction function, allowing users to dynamically set the starting point and endpoint by clicking the map, improving user experience and simulating the path planning needs in actual surgery;

(6) Path planning algorithm: The A* algorithm implemented by MATLAB was used for path planning. The algorithm considered the spatial characteristics of blood vessel images and optimized the path search efficiency by setting the heuristic function as the Manhattan distance. For each step of the neighbor search, the possible path selection of the surrounding eight directions was realized, ensuring that the selected path did not cross the image boundary or obstacle, and found the shortest path from the beginning to the end;

(7) Parameter setting: The standard deviation of the Gaussian filter in image processing was set to 2.0, and the threshold of edge detection was set to 11. In the A* algorithm, the heuristic function was calculated based on the Manhattan distance, and algorithm parameters were configured, such as whether to display the exploration process.

B. Experimental Results and Analysis

The experimental results were displayed, primarily including path visualization: the shortest path found by the algorithm was displayed, and the red line was used to depict the shortest path from the starting point to the end point. When the algorithm determined the end point, it was formed by tracing the parent node line. The red line intuitively represented the optimal solution found by the algorithm, namely the shortest path. The green area was used to mark the nodes that the algorithm was currently examining in real time. This marking helped the observer track the algorithm's progress and understand the direction and area it was searching on the map. The blue area represented the viable neighbors of the current node, that is, those nodes that had been added to the open list (to be explored list) and were waiting to be evaluated, and it highlighted the diffusion process of the algorithm, that is, how to explore possible paths around the current node.

The second aspect was obstacle processing: obstacles on the map (areas and boundaries represented by black lines) were successfully avoided by the algorithm, proving the effectiveness of the A* algorithm in dealing with obstacles. User interaction: Displayed an interactive interface where the user selected the starting point and the end point, and highlighted the location of the starting point (red asterisk) and the end point (red asterisk) on the map, emphasizing the interactivity and flexibility of the experiment.

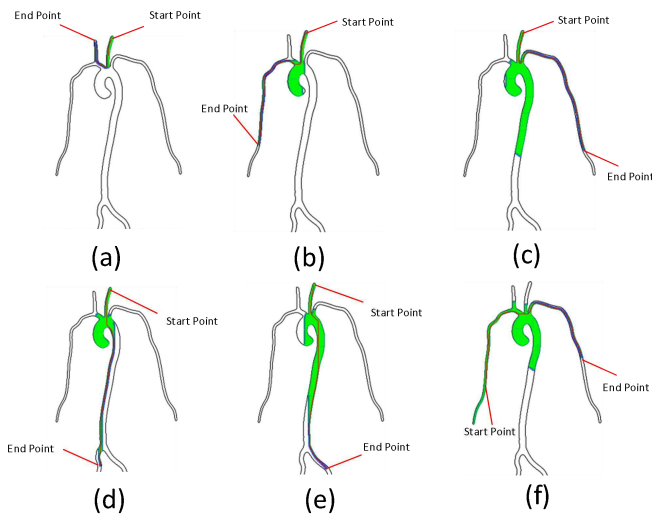


Fig. 5 Experimental results of different starting and ending points to determine the vascular path.

TABLE I RESULTS OF 6 GROUPS OF EXPERIMENTAL DATA		
	Number of Pixels	Time (s)
Fig. 5 (a)	120	5.24
Fig. 5 (b)	274	63.86

Fig. 5 (c)	336	251.07
Fig. 5 (d)	378	106.91
Fig. 5 (e)	388	186.65
Fig. 5 (f)	389	221.70

The experiment determined six pairs of different starting and ending points on vascular images to obtain the shortest path results as shown in Fig. 5. Table I presented the length of the shortest paths found in the six sets of experiments, as well as the time expended. It was observable that experiments Fig. 5 (a) and (b) required a shorter amount of time to find the shortest paths, and the paths themselves were shorter in length. In contrast, the remaining images required a longer time to determine the shortest paths, which were also longer in length. The discrepancies were primarily due to the following reasons: Complexity of the path and obstacles: In the subsequent images, the paths between the starting and ending points became increasingly complex, and the number of obstacles (such as vascular curves, bifurcations, or narrow areas) grew, necessitating the A* algorithm to evaluate more nodes in order to bypass these impediments and locate a viable path to the destination. This directly prolonged the time taken to calculate the path. The image's distance between the starting and ending points: As the experiment progressed, the image's distance between the starting and ending points incrementally increased, which directly led to the algorithm needing to assess more nodes to find the shortest path between the two points. The efficiency of the algorithm's search: Given the potential irregular shapes and tortuous paths in vascular images, the Manhattan distance might not be the most accurate method for assessing the actual cost of movement, potentially causing the algorithm to explore more unnecessary nodes before reaching the target.

The experimental results were analyzed, first of all, the path effectiveness: The experimental results proved that the A* algorithm could effectively find the shortest path in the complex environment containing obstacles. The path between the starting point and the end point obviously avoided obstacles, which showed the practical application value of the algorithm. The second aspect was performance analysis: the running time of the algorithm and the number of nodes explored reflected its performance. In this experiment, when the structure of the blood vessel was not complicated (no bifurcation, no significant change in blood vessel diameter), the A* algorithm showed high efficiency and could find the shortest path in a short time. However, when the structure of the blood vessel was complicated (multiple bifurcations, sharp curvature, or significant changes in blood vessel diameter), the A* algorithm showed low efficiency. It took a long time to determine the shortest path of the blood vessels. The function of the heuristic function: By adjusting the heuristic function (such as Manhattan distance), the search efficiency and path optimization of the algorithm could be affected. The heuristic function used in this experiment was $f(x) = g(x) + h(x)$, where h was calculated based on the Manhattan distance, and this choice performed well in balancing search speed and accuracy.

In conclusion, this experiment not only validated the effectiveness of the A* algorithm in path planning but also made every step of the algorithm intuitive and understandable through visual means. Future work can explore the effects of different heuristic functions or apply the algorithm to more complex real-world scenarios.

V. CONCLUSIONS AND FUTURE WORK

This paper experimentally verified the effectiveness and practicality of the A* algorithm in path planning applications within the vascular interventional robot system. By preprocessing vascular CTA images and calibrating obstacles, the experiment demonstrated that the A* algorithm could find the shortest path from a given starting point to the endpoint. Furthermore, the user interaction design in the experiment also validated its potential application value in real clinical environments, providing the surgical team with an intuitive operating interface. This research deepened the understanding of the application of the A* algorithm in real medical scenarios.

The future will include exploring algorithm optimization, validating the algorithm in real clinical settings, and developing integrated applications for comprehensive decision support in vascular path planning.

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